

Part 1: Project Documentation & DAX Measures

In this project, a dedicated Time Dimension (Calendar Table) was created, and essential analytical metrics were built using DAX (Data Analysis Expressions) to ensure accurate, dynamic, and interactive calculations:

1. Calendar Table (Time Dimension)

A separate calendar table was generated to establish proper date relationships and enable hierarchical drill-down capabilities across years and months:

Calendar = CALENDAR(MIN('Sample - Superstore'[Order Date]), MAX('Sample - Superstore'[Order Date]))

Extracted Calculated Columns:

Year = YEAR('Calendar'[Date])

Month = MONTH('Calendar'[Date])

Month Name = FORMAT('Calendar'[Date], "MMM")

2. Financial Key Performance Indicators (KPIs)

Total Sales:

Total Sales = SUM('Sample - Superstore'[Sales])

Total Profit:

Total Profit = SUM('Sample - Superstore'[Profit])

Profit Margin %: *(Formatted as a percentage %)* •

Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)

3. Operational & Customer Behavior Metrics

Total Orders (Unique Invoices):

Total Orders = DISTINCTCOUNT('Sample - Superstore'[Order ID])

Total Quantity Sold:

Total Quantity = SUM('Sample - Superstore'[Quantity])

Average Discount Rate: •

Avg Discount = AVERAGE('Sample - Superstore'[Discount])

Part 2: Business Insights Report

Based on the interactive two-page Power BI dashboard, the following data-driven insights have been uncovered to support strategic business decisions:

1. Page 1: Financial Performance (Executive Summary)

Timeline Growth Trend: The company shows consistent and steady year-over-year growth in both sales and profits from 2014 to 2017, indicating successful market expansion and increasing market share.

Main Category Performance: The Technology sector drives the highest revenue (\$0.41M), closely followed by the Furniture sector (\$0.39M).

The Root Problem (Hidden Leaks): Although the Furniture category generates massive revenue, the sub-category of **Tables** incurs heavy losses, dropping to nearly **-\$9.7K**. This is the primary driver of profit bleeding within the entire company, followed closely by **Bookcases**.

Geographical Insights: California and New York serve as the core profit engines for the store, showcasing high sales and strong margins. Conversely, Texas represents a major red zone, generating substantial losses despite high sales volumes.

2. Page 2: Customer Operations (Customer & Shipping Insights)

Customer Segment Power: The Individual **Consumer** segment is the primary engine of the business, accounting for approximately **45%** of total sales and profits, followed by the **Corporate** segment at **27%**.

Logistics & Shipping Behavior:

Standard Class shipping is by far the most preferred method across all segments, exceeding **3K unique orders**.

Consumers utilize every shipping option more frequently than other segments, including premium options like Same Day shipping, highlighting the urgency of individual buyers compared to businesses that opt for economical shipping.

Product-Customer Cross Analysis (Matrix Insights):

Phones and **Chairs** are the top-performing products across all customer segments, with sales for each exceeding \$320K.

The detailed breakdown matrix confirms that losses in the **Tables** sub-category are consistently repeated across *all* segments (Consumer, Corporate, and Home Office). This proves that the issue is not tied to a specific customer type, but rather to **flawed product pricing or excessive discount policies**.

Strategic Business Recommendations:

Optimize or Halt Discounts on Tables: The discount strategy for Tables must be audited immediately. Data from both Python and Power BI shows that aggressive discounts (reaching up to 80%) are directly turning high sales volumes into major losses.

Double Down on High-Margin Products: Marketing campaigns should aggressively prioritize **Phones** and **Chairs**, as they demonstrate stable, high-profit yields across all business segments.

Audit the Texas Market: A deeper business review is needed for the Texas region to investigate local operational losses and replicate the highly profitable models utilized in California and New York.